Transmission Lines

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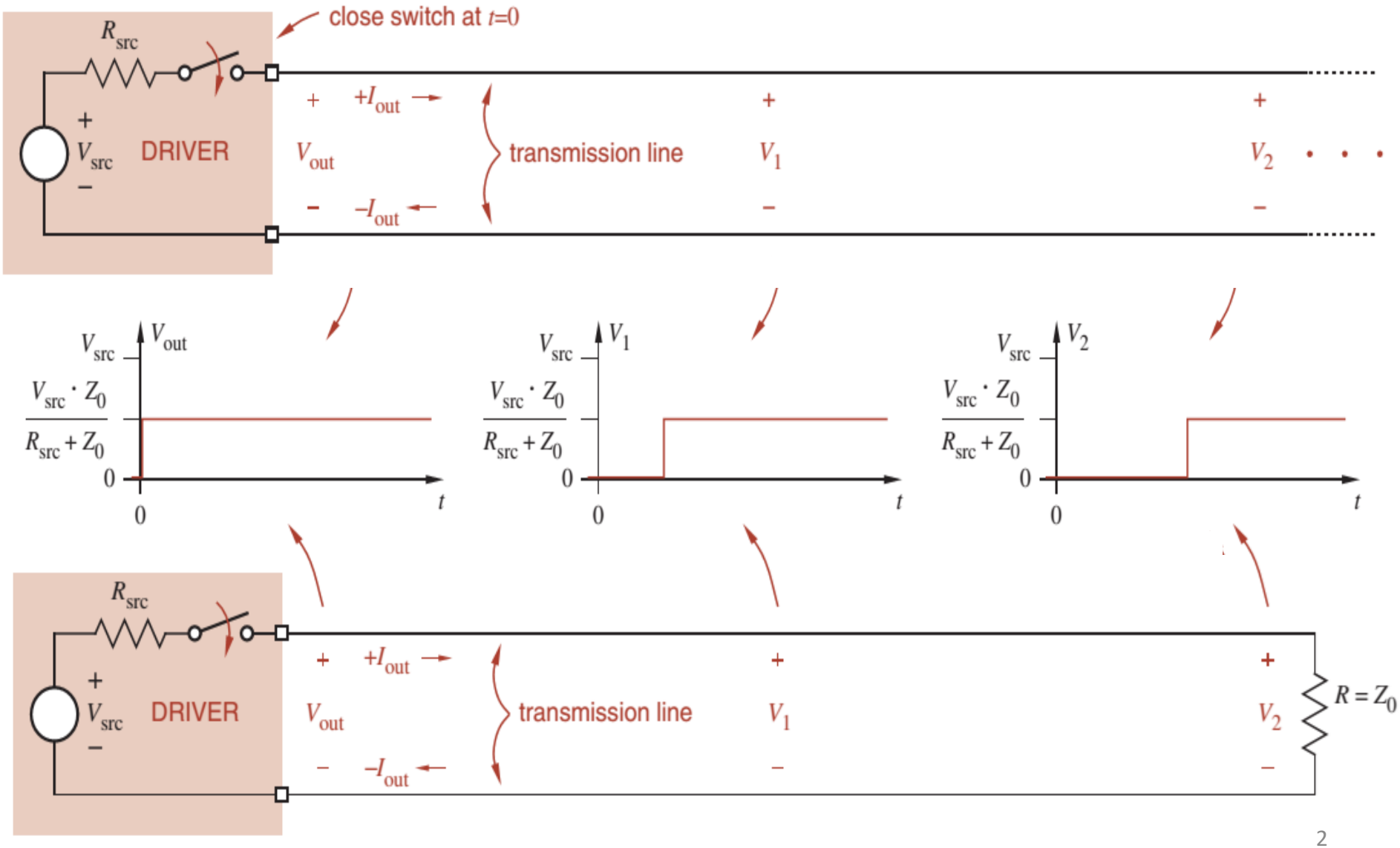
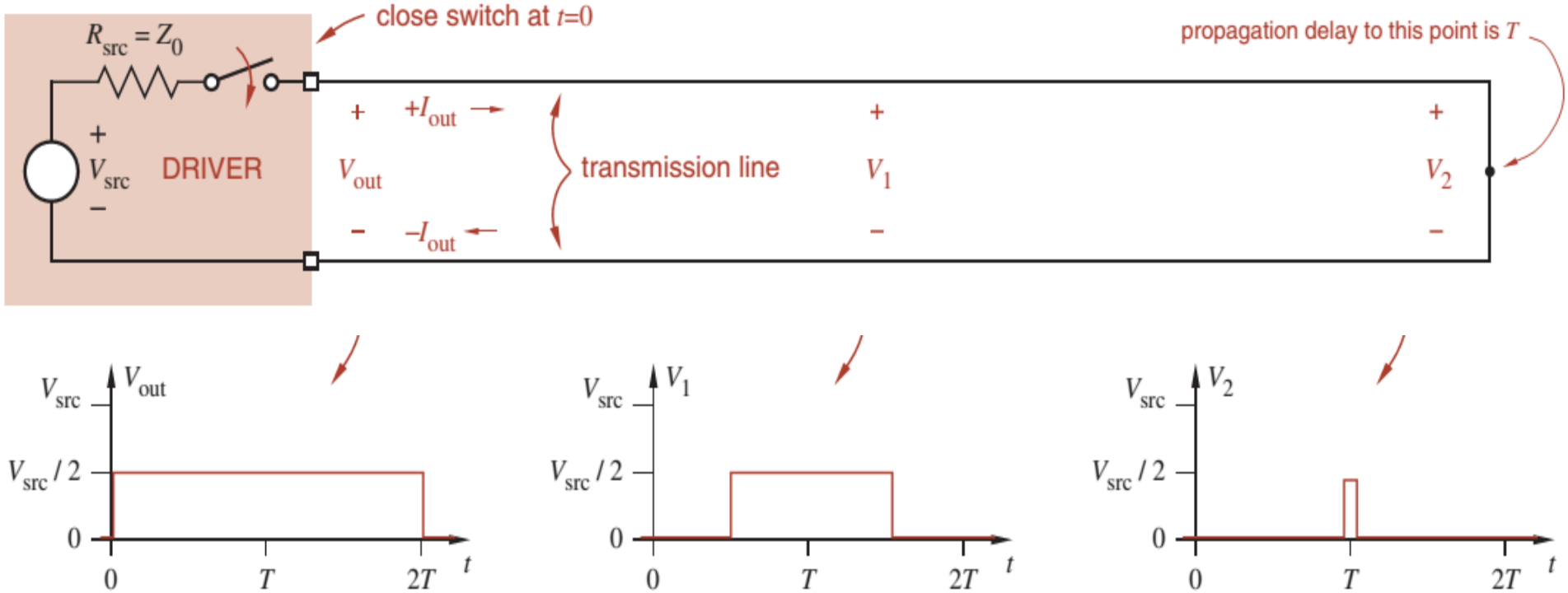
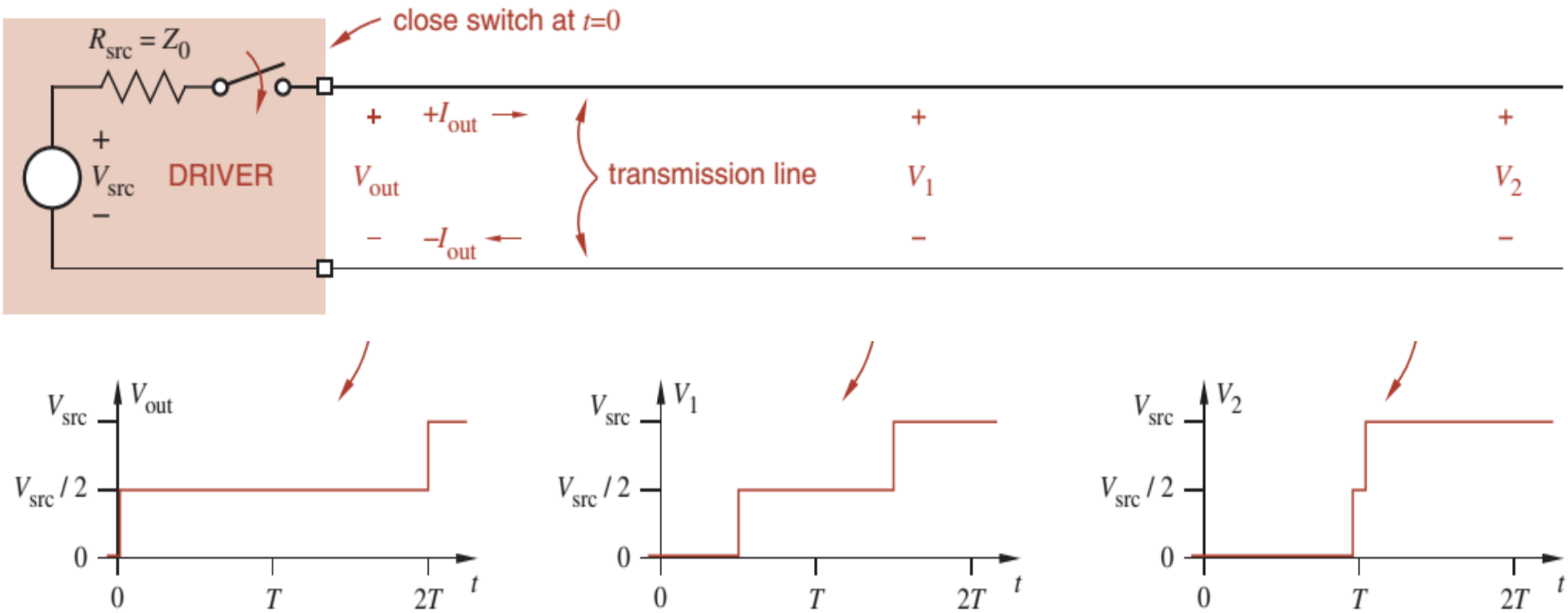
Infinite Length Transmission Line



Short Circuited Finite Length Transmission Line



Open Circuited Finite Length Transmission Line

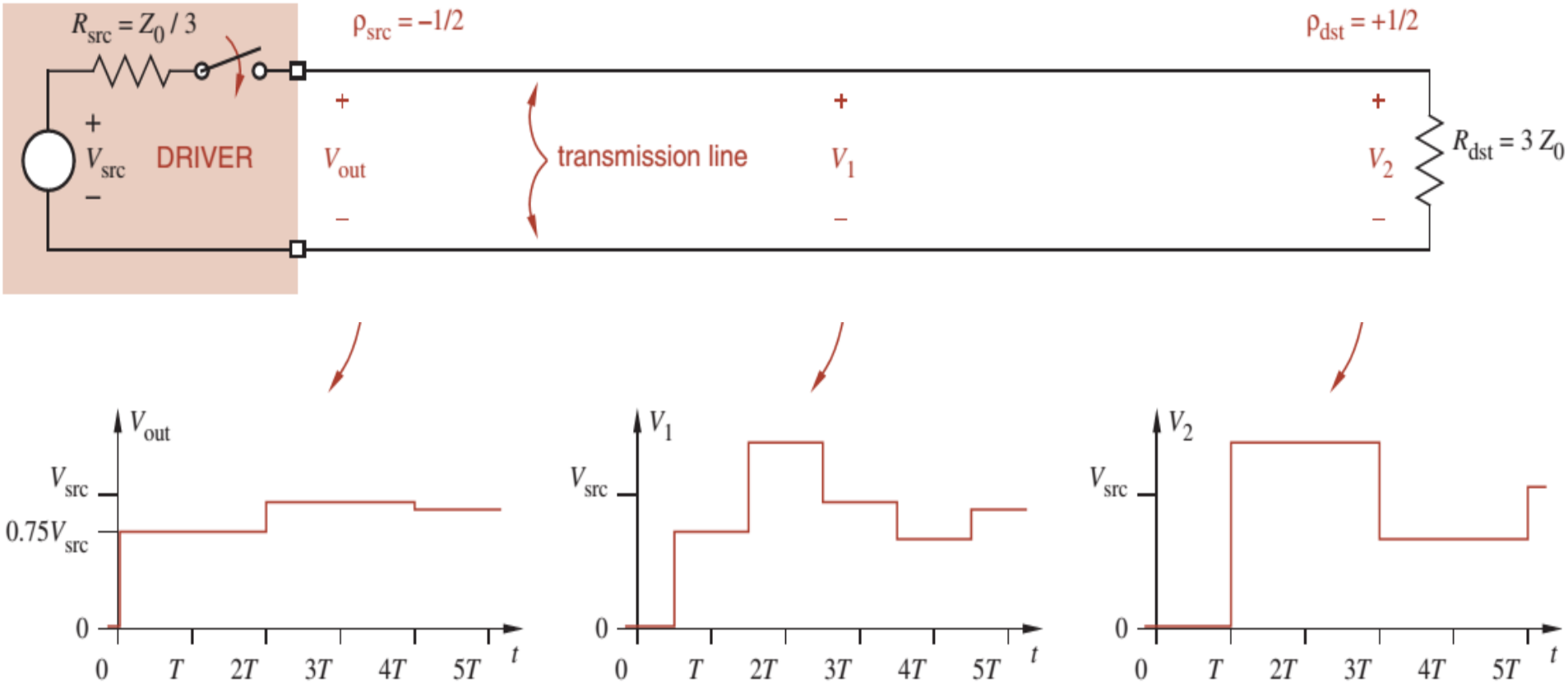


Reflection Coefficient (ρ)

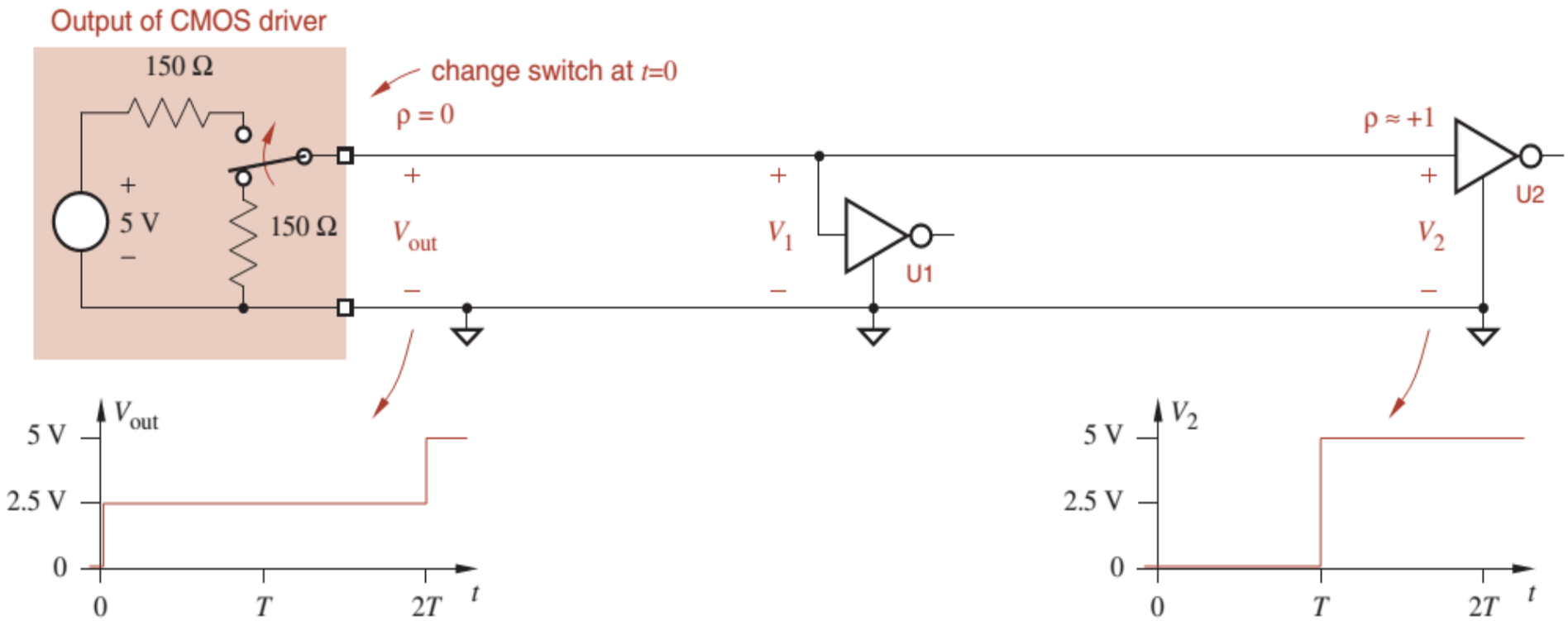
$$\rho = \frac{Z_{\text{term}} - Z_0}{Z_{\text{term}} + Z_0}$$

- Well Matched Transmission Line: $Z_{term} = Z_0 \rightarrow \rho = 0$
- Short Circuited Transmission Line: $Z_{term} = 0 \rightarrow \rho = -1$
- Open Circuited Transmission Line: $Z_{term} = \infty \rightarrow \rho = +1$

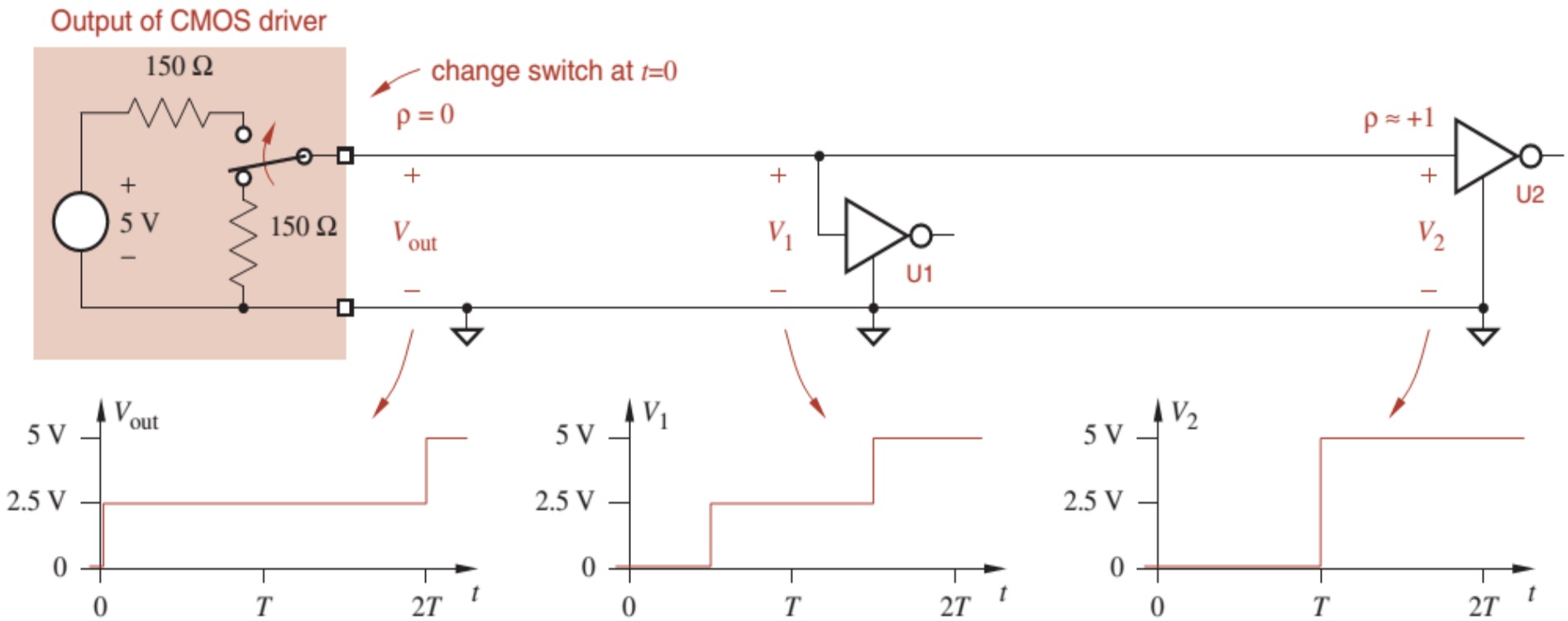
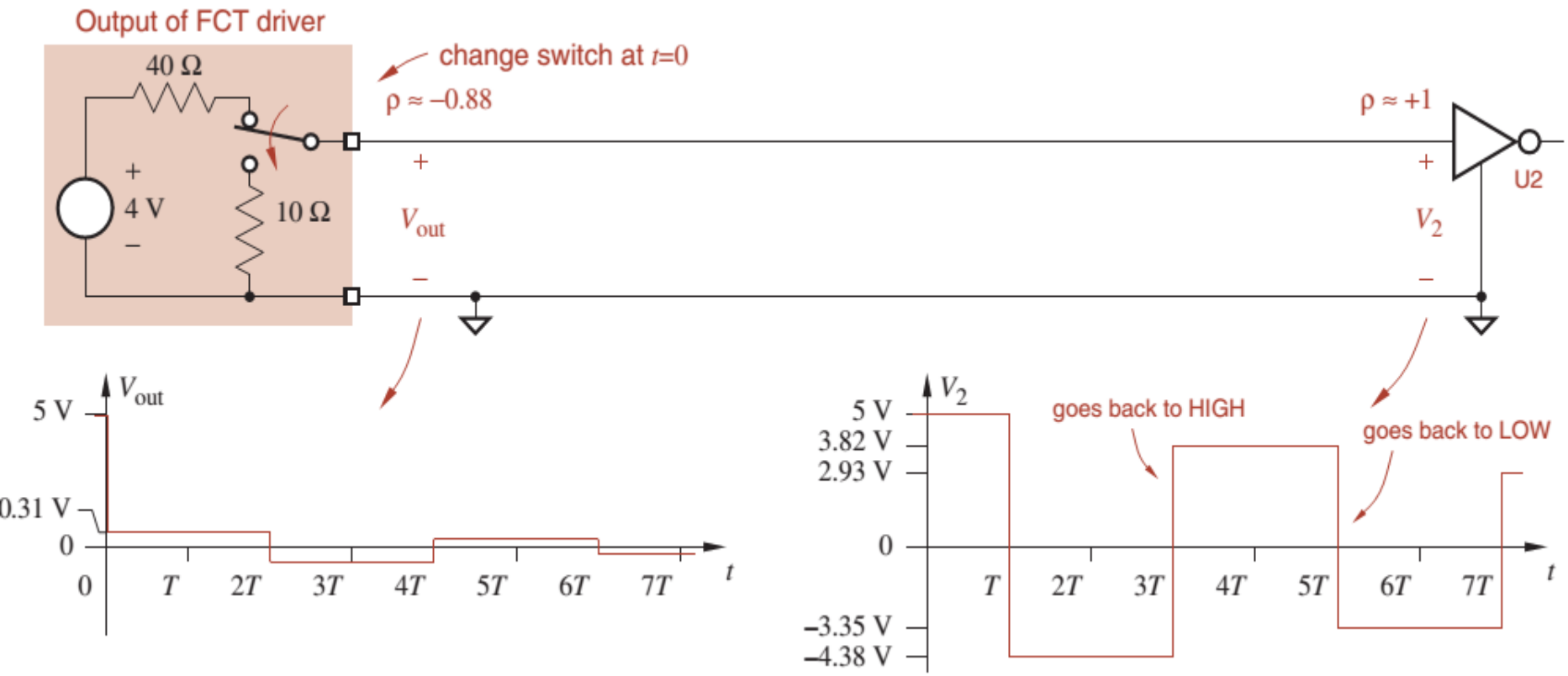
Non-Matched Transmission Line, Overshoot



Transmission Line (Side) Effect



Transmission Line (Side) Effect



Transmission Line (Side) Effect, Undershoot, Ringing